

Simulation study: mixture of experts model

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 3$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.35, 0.40, 0.25)$
- Degrees of freedom $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution $\Sigma_1, \Sigma_2, \Sigma_3$
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(mewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.35, 0.40, 0.25),
  nus = c(8, 12, 3),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 3

# run a mewishart model
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- mewishart(S_list,
  K = my_K_intial, init_pi = c(0.45, 0.40, 0.15), mh_sigma = 1,
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)

Iter    1 | LL=-697.3 | 62.3 iter/sec | acc_rate_nu=[0.000 0.000]
Iter  500 | LL=-633.1 | 868.4 iter/sec | acc_rate_nu=[0.028 0.038]
Iter 1000 | LL=-637.9 | 891.9 iter/sec | acc_rate_nu=[0.023 0.037]
Iter 1500 | LL=-628.9 | 901.1 iter/sec | acc_rate_nu=[0.023 0.033]
Iter 2000 | LL=-630.0 | 910.7 iter/sec | acc_rate_nu=[0.021 0.033]
```

```

Iter 2500 | LL=-631.9 | 910.9 iter/sec | acc_rate_nu=[0.022 0.032]
Iter 3000 | LL=-635.4 | 914.8 iter/sec | acc_rate_nu=[0.022 0.031]
Iter 3500 | LL=-630.7 | 916.7 iter/sec | acc_rate_nu=[0.022 0.034]
Iter 4000 | LL=-631.8 | 919.9 iter/sec | acc_rate_nu=[0.022 0.034]
Iter 4500 | LL=-633.0 | 918.4 iter/sec | acc_rate_nu=[0.022 0.034]
Iter 5000 | LL=-634.4 | 920.1 iter/sec | acc_rate_nu=[0.022 0.035]
Iter 5500 | LL=-632.7 | 918.7 iter/sec | acc_rate_nu=[0.021 0.035]
Iter 6000 | LL=-630.9 | 918.0 iter/sec | acc_rate_nu=[0.020 0.036]
Iter 6500 | LL=-635.9 | 918.8 iter/sec | acc_rate_nu=[0.020 0.036]
Iter 7000 | LL=-631.8 | 919.9 iter/sec | acc_rate_nu=[0.020 0.036]
Iter 7500 | LL=-632.5 | 918.3 iter/sec | acc_rate_nu=[0.021 0.037]
Iter 8000 | LL=-633.6 | 917.9 iter/sec | acc_rate_nu=[0.020 0.037]
Iter 8500 | LL=-630.9 | 917.4 iter/sec | acc_rate_nu=[0.020 0.037]
Iter 9000 | LL=-635.4 | 916.5 iter/sec | acc_rate_nu=[0.021 0.037]
Iter 9500 | LL=-635.6 | 915.9 iter/sec | acc_rate_nu=[0.021 0.037]
Iter 10000 | LL=-631.7 | 915.3 iter/sec | acc_rate_nu=[0.021 0.038]

```

```

end_time <- Sys.time()
elapsed_time <- end_time - start_time
elapsed_time

```

Time difference of 10.93552 secs

```

# posterior summaries
colMeans(fit$pi) # estimated pi

```

```
[1] 0.3444463 0.3451983 0.3103554
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit$nu, 2, mean) # estimated nu
```

```
[1] 3.551723 15.413261 7.798432
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```

# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean[, , 1])

```

```

      [,1]      [,2]
[1,] 3.2234426 0.3320974
[2,] 0.3320974 2.8227677

```

```
print(sigma_posterior_mean[, , 2])
```

```

      [,1]      [,2]
[1,] 1.6349611 0.3731719
[2,] 0.3731719 1.1678878

```

```
print(sigma_posterior_mean[, , 3])
```

```

      [,1]      [,2]
[1,] 0.4937036 0.2408740
[2,] 0.2408740 0.7207908

```

```
as.data.frame(Sigma_list)
```

```
      X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2  2.0  0.6  4.0  0.2
2 0.2 0.7  0.6  1.5  0.2  3.0
```

```
(a1 <- apply(sigma_posterior_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa
```


[1] 0.669352170 0.346453873 0.003813278

Check identifiability issue

```
# re-run the model with wrong initial weights to check identifiability issue
set.seed(1234)
fit2 <- mewishart(S_list,
  K = my_K_intial, init_pi = c(0.15, 0.40, 0.45), mh_sigma = 1,
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter    1 | LL=-697.3 | 930.2 iter/sec | acc_rate_nu=[0.000 0.000]
Iter   500 | LL=-641.6 | 891.1 iter/sec | acc_rate_nu=[0.032 0.042]
Iter  1000 | LL=-630.4 | 906.1 iter/sec | acc_rate_nu=[0.032 0.037]
Iter  1500 | LL=-630.5 | 916.0 iter/sec | acc_rate_nu=[0.034 0.037]
Iter  2000 | LL=-631.5 | 912.7 iter/sec | acc_rate_nu=[0.035 0.039]
Iter  2500 | LL=-634.2 | 920.5 iter/sec | acc_rate_nu=[0.032 0.038]
Iter  3000 | LL=-631.6 | 919.6 iter/sec | acc_rate_nu=[0.029 0.038]
Iter  3500 | LL=-632.9 | 923.2 iter/sec | acc_rate_nu=[0.028 0.039]
Iter  4000 | LL=-630.2 | 927.0 iter/sec | acc_rate_nu=[0.029 0.040]
Iter  4500 | LL=-631.4 | 929.5 iter/sec | acc_rate_nu=[0.028 0.040]
Iter  5000 | LL=-633.5 | 929.6 iter/sec | acc_rate_nu=[0.027 0.039]
Iter  5500 | LL=-630.6 | 928.8 iter/sec | acc_rate_nu=[0.027 0.038]
Iter  6000 | LL=-634.8 | 928.5 iter/sec | acc_rate_nu=[0.027 0.038]
Iter  6500 | LL=-631.1 | 927.6 iter/sec | acc_rate_nu=[0.027 0.038]
Iter  7000 | LL=-630.7 | 925.5 iter/sec | acc_rate_nu=[0.027 0.038]
Iter  7500 | LL=-633.3 | 923.4 iter/sec | acc_rate_nu=[0.026 0.038]
Iter  8000 | LL=-634.3 | 923.3 iter/sec | acc_rate_nu=[0.025 0.038]
Iter  8500 | LL=-634.3 | 921.8 iter/sec | acc_rate_nu=[0.025 0.038]
Iter  9000 | LL=-632.8 | 923.3 iter/sec | acc_rate_nu=[0.025 0.038]
Iter  9500 | LL=-630.7 | 921.7 iter/sec | acc_rate_nu=[0.025 0.037]
Iter 10000 | LL=-630.3 | 920.5 iter/sec | acc_rate_nu=[0.025 0.037]
```

```
colMeans(fit2$pi) # estimated pi
```

```
[1] 0.3180408 0.3408005 0.3411587
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit2$nu, 2, mean) # estimated nu
```

```
[1] 7.173981 15.907656 3.448413
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```

# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean2 <- Reduce("+", fit2$Sigma) / length(fit2$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean2[, , 1])

```

```

      [,1]      [,2]
[1,] 0.5423603 0.2577193
[2,] 0.2577193 0.7860789

```

```

print(sigma_posterior_mean2[, , 2])

```

```

      [,1]      [,2]
[1,] 1.5715962 0.3554456
[2,] 0.3554456 1.1214144

```

```

print(sigma_posterior_mean2[, , 3])

```

```

      [,1]      [,2]
[1,] 3.4212245 0.3795643
[2,] 0.3795643 2.9794429

```

```

as.data.frame(Sigma_list)

```

```

  X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2  2.0  0.6  4.0  0.2
2 0.2 0.7  0.6  1.5  0.2  3.0

```